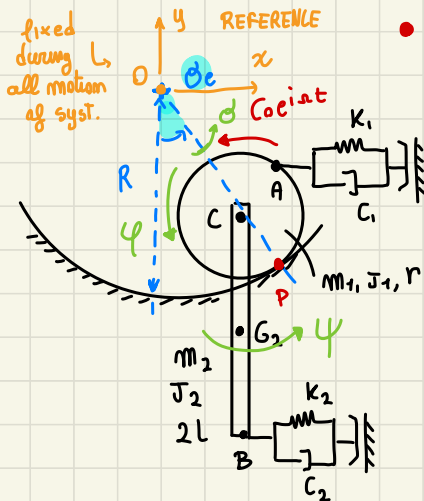




ESE 12 PREPARATION FOR PROBLEM 1 (EXAM)

• Exam MIDTERM 2021



DISK rolling (without sliding) on a circular guide
concave + Rigid bar hinged to the center of
disk (2 bodies)

DOF

6 DOF

• HINGE (C) - 2 DOF

• ROLLING W.S - 2 DOF (P)

⇒ 2 DOF

• disk motion? through
2 possible choices

2 INDIP
cohort

1) θ as measured

respect to assumption radial position

2) ψ DISK ROTATION

related in between, then we will connect it
relationship

We take ψ as
indip cohort

• Bar rotation ψ

↓

$[m_{ph}], [k_{ph}], [c_{ph}]$ as

usual, we should find $[\lambda_m], [\lambda_k], [\lambda_c], [\lambda_f]$

and $[K]_{e_{x, \psi}, g}$

$$\underline{x} = \begin{Bmatrix} \psi \\ \psi \end{Bmatrix} \quad \left[\begin{array}{l} \text{if } \psi, \psi = 0 \text{ FROZEN system} \\ \text{STILL DISK! also BAR!} \\ \text{Set of indep cohort} \end{array} \right]$$

same expression, parallel spring / damper

from TEXT DATA

$$\Delta \theta_{02} = 0$$

(in fact @ equilibrium the pendulum should
be rotate) → consistent

regardless of $\Delta \theta_{02}$ relation ship

$[K]_{e_{x, \psi}, g} = [0]$ because $\Delta \theta_{02} = 0 \rightarrow$ we don't care to find that relationship

we just need

for spring 2

$k_{g1}, k_{g2}, k_{e_{x, \psi}}$

$$[\lambda_m] = \begin{bmatrix} \dot{\varphi} & \dot{\psi} \\ -r \cos(\vartheta_e) & 0 \\ -r \sin(\vartheta_e) & 0 \\ 1 & 0 \\ -r \cos(\vartheta_e) & L \\ -r \sin(\vartheta_e) & 0 \\ 0 & 1 \end{bmatrix} \begin{matrix} V_{Cx} \\ V_{Cy} \\ \omega_1 \\ V_{G_{2x}} \\ V_{G_{2y}} \\ \omega_2 \end{matrix}$$

We choose a **reference 0** fixed
otherwise we have issue on
relative motion CAREFULL

We compute $[\lambda_m]$ by

1 • WRITING the complete
NON LIN EXP and expand
taking only 1st ORD

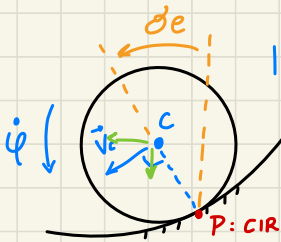
2 • super-imposition approach

→ compute by column
by RIVALS THEOREM

Considering $\dot{\varphi} \neq 0, \dot{\psi} = 0$ / $\dot{\varphi} = 0, \dot{\psi} \neq 0$

a) $\dot{\varphi} \neq 0, \dot{\psi} = 0$

$\omega_2 = 0$ for sure NOT ROTATING Bar and any point in bar moves as C
so findy $V_{Cx} = V_{G_{2x}}$, $V_{Cy} = V_{G_{2y}}$ We just need to describe C motion

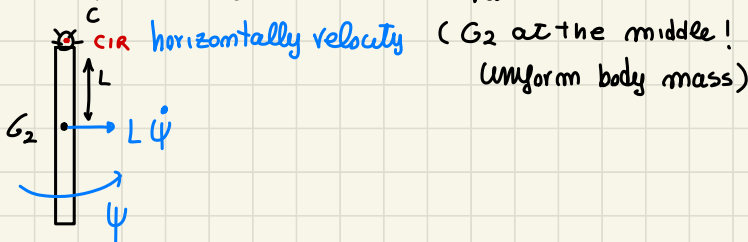


$$|\vec{V}_C| = r \dot{\varphi} \quad V_{Cx} = (r \cos(\vartheta_e)) \dot{\varphi} = V_{G_{2x}}$$

moving opposite
to the choice

$\omega_1 = \dot{\varphi}$ directly it is the absolute ROTATION

b) $\dot{\varphi} = 0, \dot{\psi} \neq 0$ DISK NOT ROTATING, the entire disk is still $\Rightarrow$ C has no velocity
 $V_C = 0, \omega_1 = 0$ while the entire bar rotates around C (its CIR)

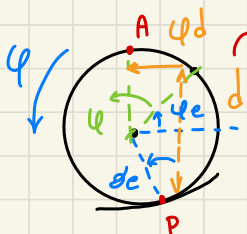


then we should find for the springs

$$\begin{bmatrix} \lambda_k \\ \parallel \\ \lambda_c \end{bmatrix} = \begin{bmatrix} r(\cos(\varphi_e) + \sin(\varphi_e)) & 0 \\ r \cos(\varphi_e) & -2L \end{bmatrix} \begin{bmatrix} \lambda_{e1} \\ \lambda_{e2} \end{bmatrix}$$

Notice A , the
- endpoint of K_{r_1}
attached to the circumference
of the disk (so distant r from c)

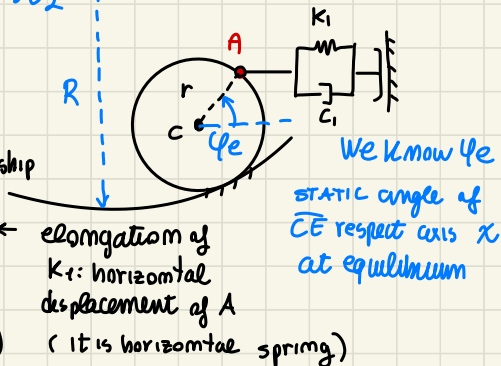
IF small φ
Rotation respect φ_e



only
vertical
displacement
affect
horiz.
displacement
(*) NEXT

So we want
a linear relationship

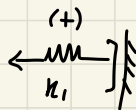
between φ and X_A (carefully to sign convention)



$d = \bar{P}A$ distance which is $PC + CA$ projected on y

$$d = r \cos \vartheta e + r \sin \varphi e$$

this produce a displacement of A to the left
so increasing length



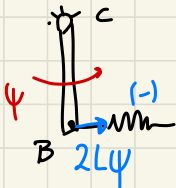
While ΔE_2 when $\psi = 0$ move as G_2 (horizontal spring) so as

$X_{G2} = X_c \rightarrow$ from $[\lambda m]$ we can take the relationship $+r \cos(\theta_e)$

moving spy on the left $\leftarrow m_{(+)}^{k_2}$

While $\varphi=0$ frozen disk $\rightarrow$ obviously $\Delta E_1=0$

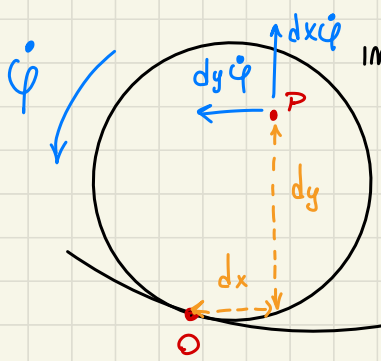
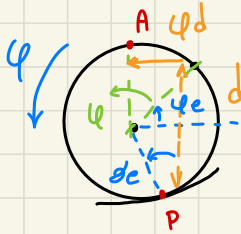
While when bar is rolling



it compress
the spring

$\psi > 0$ decrease
 ΔE_2

(*) NOTE



In general, when:

body rotates
around $O \in C.I.R$

distance between O, P

movement of P

respect displacement

only dy affect

x_P direction!

So we can solve this

problem by considering only

vertical $\overline{PA}$ distance d (even if not on same line)

for $[\lambda f]$ we just need infinitesimal virtual disk rotation $\delta\varphi$
 where Coe^{int} act

$$[\lambda f] = \begin{bmatrix} 1 & 0 \end{bmatrix} \delta\varphi$$

↑
independent!

no active forces
 on row 2 of equation
 (respect φ)

II ORD K_{el}, K_g , so

by II ord power series expansion of $y_c, y_{o2}, \Delta\theta_1$

imageneous displacement:

$$y_c = -\bar{R} \cos(\theta_e + \sigma) = \bar{R} = (R-r)$$

(-) sign because C below origin 0

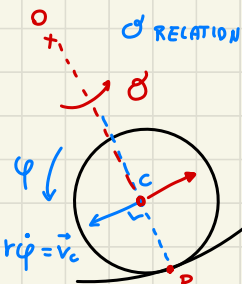
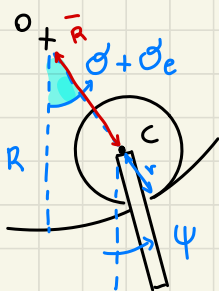
II ORD

$$\approx -\bar{R} \cos(\theta_e) \left(1 - \frac{\theta_e^2}{2}\right) +$$

$$+\bar{R} \sin(\theta_e) \cdot \sigma =$$

{ vary with position, we
 should express σ respect φ !
 (NOT an indep coord)

σ RELATIONSHIP with φ ?



C as movement
 circular around O

for $\sigma > 0$

$$\vec{v}_c = \vec{CO} \dot{\sigma} = \bar{R} \dot{\sigma}$$

same DIRECTION
 $-r\dot{\varphi} = +\bar{R}\dot{\sigma}$
 they must be the SAME! Independently

INTEGRAT. from P.O.V

$$\dot{\sigma} = -\frac{r}{\bar{R}} \dot{\varphi} \Rightarrow \sigma = -\frac{r}{\bar{R}} \varphi + \text{constant}$$

↑
CONSTANT

σ, φ as increasing angle
 $\varphi = 0$ correspond to $\sigma = 0$
 so constant = 0

$$y_c = -\bar{R} \cos(\theta_e) + \bar{R} \sin(\theta_e) \cdot \left(-\frac{r}{\bar{R}}\right) \varphi$$

CONSTANT

$$+ \bar{R} \cos(\theta_e) \frac{r^2}{\bar{R}^2} \frac{\varphi^2}{2} =$$

$$= -r \sin(\theta_e) \varphi + \frac{r^2}{\bar{R}} \cos(\theta_e) \frac{\varphi^2}{2}$$

same as
 found in
 V_{cy} respect φ
 on $[\lambda m(2.1)]$

II ORD
 TERM

$$[K_{g1}] = m_1 g \begin{bmatrix} \frac{r^2}{\bar{R}} \cos(\theta_e) \\ 0 \end{bmatrix}$$

no cross
 product

no $\sigma^2/2$
 terms on y_c

While to write $y_{G2} = y_c - L \cos(\psi) \approx -r \sin(\theta_e) \psi + \frac{r^2}{R} \cos(\theta_e) \frac{\psi^2}{2} - L \left(\cancel{1} - \frac{\psi^2}{2} \right)$

$\uparrow$
 neglecting
 ORDER
 OM y_c

CONST

$$[K_{g2}] = m_2 g \begin{bmatrix} \frac{r^2}{R} \cos(\theta_e) & 0 \\ 0 & L \end{bmatrix}$$

no cross product

then we need to find an expression for Δe_1 @ I ORD terms expansion
 k_1 depends only on disk motion $\rightarrow [K_{el II,1}]$

$$\Delta e_1 = -x_A \rightarrow$$

measuring x_A
 on the right,
 compressing the spring!

expressing θ respect ψ (our indep coh)

without approx
 $\downarrow$

$$x_A = \bar{R} \sin(\theta_e + \theta) + r \cos(\psi_e + \psi) =$$

$$\approx \bar{R} \cos(\theta_e) \theta + \bar{R} \sin(\theta_e) \left(\cancel{1} - \frac{\theta^2}{2} \right)$$

CONST

$$+ r \cos(\theta_e) \left(\cancel{1} - \frac{\psi^2}{2} \right) - r \sin(\psi_e) \psi$$

CONST

$$x_A = -\bar{R} \cos(\theta_e) \frac{r}{R} \psi - \bar{R} \sin(\theta_e) \frac{r^2}{R^2} \frac{\psi^2}{2} - r \cos(\theta_e) \frac{\psi^2}{2} - r \sin(\psi_e) \psi$$

$\downarrow$

$$\Delta e_1 = -x_A = \underbrace{\left(r \cos(\theta_e) + r \sin(\psi_e) \right) \psi}_{\text{I ORD as found on } [K_1] \text{ respect } \Delta e_1} + \underbrace{\left(\frac{r^2}{R} \sin(\theta_e) + r \cos(\theta_e) \right) \frac{\psi^2}{2}}_{\text{II ORD term to write}}$$

$$[K_{el II,1}] = k_1 \Delta e_{\theta_1} \begin{bmatrix} \frac{r^2}{R} \sin(\theta_e) + r \cos(\theta_e) & 0 \\ 0 & 0 \end{bmatrix}$$

IF $[K] < 0 \Rightarrow$ neg coefficient $[K(1,1)] \rightarrow$ INSTABILITY! so ω, X don't exist
 overall as $\Sigma [K]$ because it is unstable,

• FRF has no meaning, never stabilize on stationary behaviour!

$[K]$ part related to indep coord $\Rightarrow [K_{FF}]$ symm and pos def
(this don't valid for $[K_{u}]$)

↑ check that
 $[K_{FF}], [K_{FF}], [C_{FF}]$ symm and
Pos. def!
both diagonal terms > 0 strictly,
check on MATLAB

If $[K] < 0$ you
get $\omega^2 \geq 0$ so
getting one zero

$\pm \omega_1, \pm i\omega_2$
Im NAT FREQ! $\rightarrow$ WRONG result!

We have found all MATRIX $\rightarrow$ MATLAB find $[M], [C], [K]$

Without partitioning, we don't have
constraint displacement!

and QUESTION 2 compute
 ω, X easily

QUESTION 3: compute FRF also using $[\lambda, f]$

finding the required output Mod/phase

- we want X_{G2} so we want to find a IORD
relationship between X_{G2} and $\varphi, \dot{\varphi}$ you can use

$[\lambda_m]$ row u relative to V_{G2x} $\begin{bmatrix} -r \cos \theta_c & L \\ \dot{\varphi} & \dot{\varphi} \end{bmatrix}$

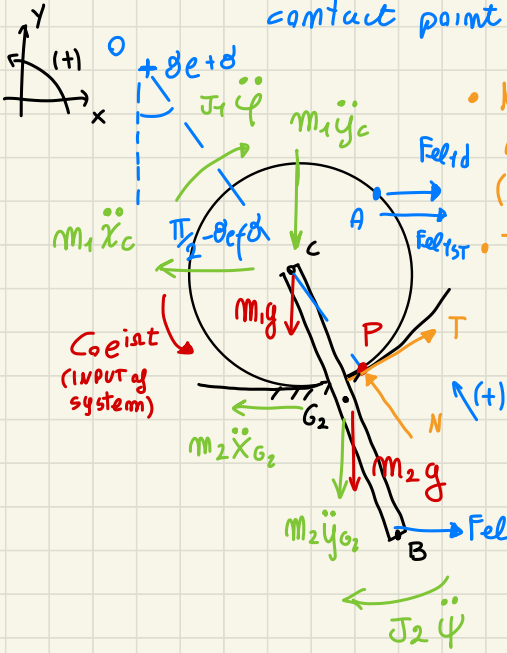
being $X = A \setminus F$ indep coord

$$\text{Output 1} = -r \cos(\theta_c) X(1) + L X(2)$$

- y_c
output 2 = $-r \sin(\theta_c) X(1)$ row on $[\lambda_m]$ relative to V_{Gy}

QUESTION 4 compute FRF applying Coe^{iat} TORQUE to The DISK
harmonic of:

output 3: tangential / normal components acting on P contact point of the disk



- $N \perp$ to the tangent on the disk contact point and guide profile (the same direction) (to contrast compenetrations)
- T parallel to tangent (created by friction usually)

↓ D'ALAMBERT principle to solve it (dym equilibrium equation)
choose wisely the approach!

F_{el} direction choosen in the direction of what happens to the spring if it get elongated

$$1. \sum_{s \in \mathcal{S}_T} F_{sN} = 0$$

sum of F parallel to N and T respect entire system

$$2. \sum_{s \in \mathcal{S}_T} F_{sT} = 0$$

general displacement $\theta_e + \theta$

1. positive θ directed like N

$$N + (m_1 \ddot{x}_c + m_2 \ddot{x}_{c_2}) \sin(\theta_e + \theta) - (m_1 g + m_2 g + m_1 \ddot{y}_c + m_2 \ddot{y}_{G_2}) \cos(\theta_e + \theta) + \cos(\frac{\pi}{2} - (\theta_e + \theta)) - (F_{el1st} + F_{el1d} + F_{el2d}) \sin(\theta_e + \theta) = 0$$

(WAGY) ↓

from here we can consider that

$$\sin(\theta_e + \theta) \sim \sin(\theta_e)$$

$$\cos(\theta_e + \theta) \sim \cos(\theta_e)$$

being θ small

so any constant term

on N lead to constant part of $N \rightarrow$ NOT relevant on FRF, mean value related just to $\omega = 0$

While FRF relate dynamic components $\rightarrow$ NO STATIC values!

We can neglect static part of $N \rightarrow$ so weights and F_{elst} can be cancel out (simplification!) (WA42)

because $\sin(\vartheta_e + \vartheta)$, $\cos(\vartheta_e + \vartheta)$ give you IORD, IORD term

$$\sin(\vartheta_e + \vartheta) \approx \sin(\vartheta_e) + \cos(\vartheta_e)\vartheta$$

IORD

$$\cos(\vartheta_e + \vartheta) \approx \cos(\vartheta_e) - \sin(\vartheta_e)\vartheta$$

and plug-in on the expression

take product and neglect IORD terms

so it survives only when multiplied by constant

while IORD sum of multiply dyn forces

$\Rightarrow$ and solve for $N = \dots$

both way are good to use $\uparrow$ WA42 accurate

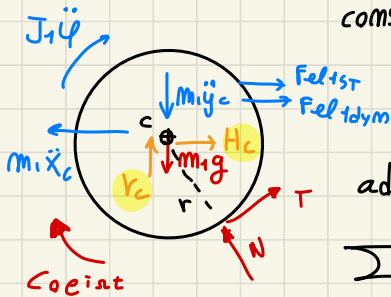
2.

$\sum_{sys} F_{//T} = 0 \rightarrow$ same expression as before but parallel to T , so $\sin \rightarrow \cos$ on that $\cos \rightarrow \sin$

OR differently

OTHERWISE

considering only the disk



also exchange ment (neglected if you consider the entire system) forces on HINGE becoming INTERNAL FORCES

additional unknowns

$$\sum_{Disk} \vec{M}_c = 0$$

both H_c, V_c has 0 am! but even $m_c g, N, m_c \ddot{y}_c, m_c \ddot{x}_c$ has 0 am!

only $\begin{cases} \bullet C_0 e^{i\omega t} \\ \bullet T \cdot r \\ \bullet J_1 \ddot{\phi} \\ \bullet H_{elst} \end{cases}$ affect the equation (equal complexity between this approach of $\sum F_{//T}$)

$$T \cdot r + C_0 e^{i\omega t} - J_1 \ddot{\phi} + F_{el_{1st}} \text{ arm} + F_{el_{1dyn}} \text{ arm}$$